Grassroots Community Currencies

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Abstract. Community currencies create liquidity within a community from the trust among its members, in units pegged by convention to the national currency and kept by an operator. Grassroots currencies let every person issue coins of their own and form liquidity by mutual credit, with no operator and no shared ledger, but their coins are undenominated and no community issues them.

Here we present *grassroots community currencies*: coins issued by a community through its treasurers, denominated by a fiat currency and backed by the treasurers' personal coins, specified by a grassroots contract and compiled from it to a smartphone app. A community is started by a person, its first treasurer; the treasurers appoint and remove treasurers by joint decision; a treasurer mints community coins; and a holder redeems a community coin for a unit of its denomination from its issuer or from any treasurer, or in kind. In the backed contract a treasurer mints only while the personal coins they hold of the other treasurers, leaving out any fraction f of them, cover their community coins in circulation. We prove that a community coin stays redeemable for one unit of its denomination while at most a fraction f of the treasurers are insolvent or removed, and show that with n treasurers each extending every other k coins of mutual credit, each can have about $(1-f)nk$ community coins in circulation. A treasurer is compensated by seigniorage, and the contract requires no fee.

The contracts are verified grassroots and compiled to a working app, whose screens we show, and a stochastic simulation of communities whose treasurers fail at random leaves no coin unredeemed while the failures stay below f.

Keywords: Grassroots community currencies · Community currencies · Mutual credit · Grassroots contracts · Backing · Grassroots platforms

1 Introduction

Community currencies create liquidity within a community from the trust among its members: Sardex [10,14] is a mutual credit system among the businesses of Sardinia, and Sarafu [15] a community inclusion currency in Kenya. Their units are pegged by convention to the national currency, and an operator keeps the ledger and the rules.

Grassroots currencies [18] let every person issue coins of their own and form liquidity by mutual credit, two persons swapping coins of their own issue, with no operator and no shared ledger; a coin is redeemable against any coin its issuer holds, and chains of such redemptions carry liquidity across a community. Their coins are undenominated. Denominated grassroots coins [23] are a person's debt

of one unit of a fiat currency, redeemable at par in kind or in that currency, and a sovereign grassroots currency is the denominated currency of a central bank.

A personal denominated coin is backed by one person, and the trust placed in a person is not the trust placed in them as a financial entity. A community coin puts the joint backing of the community's treasurers in its place.

Here we present *grassroots community currencies*: coins issued by a community, denominated by a fiat currency and backed by the personal coins of the community's treasurers. They are specified by a grassroots contract [7], a social contract that runs, written as clauses its parties undertake towards one another and rendered as speech-act schemas, from which the app is compiled down the abstraction cascade [11] to Grassroots Logic Programs [20] and to a smartphone. Section 2 presents the contracts. A community is started by a person, its first treasurer, who gives it a name and a denomination; the treasurers appoint and remove treasurers by joint decision; a treasurer mints community coins, which circulate as the coins of grassroots currencies do; and a holder redeems a community coin for a unit of its denomination from its issuer or from any treasurer, or in kind against any coin a treasurer holds. In the *backed* contract a treasurer mints only while the personal coins they hold of the other treasurers, denominated as the community's, cover their community coins in circulation after leaving out any fraction f of those treasurers, and no act may leave them uncovered.

Section 3 proves that a community coin in circulation is redeemable for one unit of its denomination while at most a fraction f of the treasurers are insolvent or removed, and shows what a community can mint: with n treasurers each extending every other k coins of mutual credit, each treasurer can have about $(1-f)nk$ community coins in circulation, $6k$ at $n = 10$ and $f = 1/3$. A treasurer is compensated by seigniorage: they pay with the community coins they mint, so the community extends them interest-free credit up to their backing, and the contract requires no fee.

Section 4 shows three applications: a cooperative of drivers, a merchants' association and a farmers' cooperative, each with a contract of its own that says who accepts the community's coins. Section 5 shows the app. The contracts are verified syntactically grassroots by the checker of [7] and compiled to a working app, whose screens are shown; the protocol realising them is grassroots [7,13], each payment is countersigned by the issuer of the coins paid [12], and a lost key or device is recovered [4]. Section 6 reports a stochastic simulation [22] of communities whose treasurers fail at random: no coin is left unredeemed while the failures stay below f, and the circulation grows with the treasurers. A community coin is a signed debt of its issuer, legally binding also in electronic form [5,24]. Section 7 reviews related work and Section 8 concludes.

2 The Contracts

GC: Grassroots Coins [18]

GC extends GKK [7].
Alice and Bob are persons.

1. ■ *Minting.* Alice may decide to issue Alice-coins.
2. ■ *Swap.* Alice and Bob, who know each other, may jointly decide to swap coins they hold.
3. ■ *Payment.* Alice may decide to pay Bob-coins they hold to Bob, whom they know.
4. ■ *Redemption.* Alice may decide to redeem a Bob-coin they hold against any coin held by Bob, whom they know.
5. ○ *Pricing.* Alice prices their offerings in Alice-coins and accepts them in payment at those prices.

GDC: Grassroots Denominated Coins

GDC extends GC [18].
Alice and Bob are persons. Dollar names one denomination, and a Dollar-denominated Alice-coin is an Alice-coin issued as Alice's debt of one Dollar.

1. ■ *Minting.* Alice may decide to issue Alice-coins denominated by a fiat currency.
2. ○ *Denomination.* A Dollar-denominated Alice-coin is Alice's debt of one Dollar, due now; Alice redeems it on presentation at face value, by a Dollar-denominated coin the bearer accepts or by one Dollar.
3. ○ *Pricing.* Alice accepts Dollar-denominated Alice-coins in payment for their offerings at one Dollar each.

GCC: Grassroots Community Coins

GCC extends GDC.
Alice, Bob, Carol and Dave are persons. A community coin is a coin of the person who minted it.

1. ■ *Community.* Alice may start a community, give it a unique name, declare its currency denomination, and is then the first treasurer of the community.
2. ■ *Treasurers.* The treasurers of a community can appoint Bob as a new treasurer by their joint decision and the consent of Bob; they can remove a treasurer Carol by a joint decision of all treasurers except Carol, with seniority as a tie breaker.
3. ■ *Minting.* While Alice is a treasurer of a community, they can mint denominated community coins, which are declared as such.
4. ○ *Fiat Redemption.* Alice may redeem a community coin they hold against a fiat coin of its denomination from the person who minted it, or from any treasurer of the community.

5. ■ *Coin Redemption.* Alice may decide to redeem a community coin they hold against any coin held by a treasurer of the community, whom they know.

GCC by itself lets a treasurer mint without bound, while every treasurer is a counterparty to a holder's redemption for fiat (GCC:4); BCC bounds minting by backing.

BCC: Backed Community Coins

BCC restricts GCC.
f is a fraction, $0 \leq f < 1$. Alice, Bob and Carol are persons. Alice's community coins *out* are those Alice minted and does not hold. Alice is *backed* in a community if, leaving out the fraction f of the treasurers appointed to it, those of whom Alice holds the most coins in its denomination, Alice holds at least as many coins of the remaining treasurers but Alice as Alice has community coins of it out. For a person removed as a treasurer of a community, the treasurers appointed to it are those appointed before they were removed.

1. ■ *Treasurers.* GCC:2, only if every treasurer of the community stays backed.
2. ■ *Minting.* GCC:3, only if Alice stays backed.
3. ■ *Holding.* GC:2, GC:3, GC:4 and GCC:5, only if every treasurer, current or removed, who gives a coin by the act stays backed.

Under GCC a holder's redemption for fiat at any treasurer is an undertaking, and a breach of it is taken to court. Under BCC the issuer of a community coin holds coins of the treasurers that cover its coins out, whichever fraction f of the treasurers fail, and a holder redeems the coin against one of a solvent treasurer by GC:4, an act the contract enforces (Theorem 3.1).

BCC restricts GCC [7, Section 8]: each of its clauses strengthens the body of an act of GCC or of GC, and changes no head. An appointment may be possible only if Bob and every treasurer extend each other additional mutual credit prior to that.

A treasurer is compensated by seigniorage: they pay with the community coins they mint, so the community extends them interest-free credit, bounded by their backing, and the contract requires no fee. A fee on redemption or interest on a loan is an instrument of grassroots bonds [21] and not a clause of this contract.

3 Backing

A treasurer is *insolvent* if they fail to fulfil a presentation of a community denominated coin, and *solvent* otherwise. A community coin is *in circulation* from its minting until it returns to its issuer. The treasurers appointed to a community are those it ever had, and for a coin of a person removed as a treasurer, those appointed before they were removed (BCC).

Theorem 3.1 (Backing). *A community coin in circulation is redeemable for one unit of its denomination while at most a fraction f of the treasurers appointed to its community are insolvent or removed.*

Proof. Let the coin be held by Alice and issued by Bob, let n be the number of treasurers appointed and I the insolvent or removed ones, $|I| \leq fn$. The coin is out, so Bob has at least one community coin out, and Bob is backed: by BCC:1 to BCC:3 every act that changes what Bob holds or owes, or the treasurers appointed, keeps Bob backed, and a removal changes neither. Leaving out the $\lfloor fn \rfloor$ treasurers of whom Bob holds the most coins, Bob holds at least one coin of the remaining ones but Bob, so leaving out the treasurers of I, who are at most as many, Bob holds a coin of a treasurer $v \notin I$, $v \neq$ Bob. By GC:4 Alice may redeem the coin from Bob against that coin of v: Bob's coins out and the coins Bob holds of the remaining treasurers each drop by at most one, the first by exactly one, so Bob stays backed and BCC:3 admits the act; it is guarded by Alice alone and enforced, so Bob, running correct code, cannot refuse it, whether Bob is solvent or not. Alice then holds a coin of v in the denomination of the community, which v, being solvent, redeems for one unit of it.

How much a community can mint. Let the community have n treasurers, and let a treasurer hold k coins of each of d other treasurers, their mutual credit in the community's denomination, and none of the rest.

Proposition 3.2 (Circulation). *A treasurer with k coins of each of $d \geq \lfloor fn \rfloor$ other treasurers is backed with up to $(d - \lfloor fn \rfloor)k$ community coins out, and no more.*

Proof. Leaving out the $\lfloor fn \rfloor$ treasurers of whom the treasurer holds the most coins leaves $d - \lfloor fn \rfloor$ of the d, of k coins each, and none of the rest.

With every treasurer a creditor of every other, $d = n-1$, a treasurer has about $(1-f)nk$ community coins out, $6k$ at $n = 10$ and $f = 1/3$, and the community about $(1-f)n^2k$; with a fraction s of the others as creditors, $d = s(n-1)$, about $(s-f)nk$ each. A new treasurer who extends every treasurer k coins of mutual credit adds about $(1-f)k$ to what each of them can have out, and can have about $(1-f)nk$ out themselves, so the circulation grows with the square of the number of treasurers.

Backing in the schemas. BCC's schemas read backing in the state before the act, with the changes the act makes to the coins Alice holds and to Alice's coins out, and write the sum in the definition of backed by Lemma 3.3, with h_v the coins of a treasurer v appointed to the community, but Alice, in its denomination that Alice holds and $L = \lfloor fn \rfloor$, as a sum over the treasurers, with no sorting; the schemas are in the repository `https://anonymous.4open.science/r/GLP-02A3`.

Lemma 3.3 (Bottom sum). *Let $h_1, \ldots, h_m \geq 0$, $0 \leq L \leq m$, and S the sum of the h_v leaving out L largest. Then $S \geq Y$ if and only if $Y \leq 0$ or $\sum_{v=1}^{m} \min(h_v, h_w) - Lh_w \geq Y$ for some w.*

Proof. Let $g(t) = \sum_{v=1}^{m} \min(h_v, t) - Lt$, and $h_{(1)} \geq \cdots \geq h_{(m)}$ the h_v in descending order. For every $t \geq 0$, $\min(h_{(j)}, t) \leq t$ for $j \leq L$ and $\min(h_{(j)}, t) \leq h_{(j)}$ for

$j > L$, so $g(t) \leq S$. If $L < m$, then $g(h_{(L+1)}) = Lh_{(L+1)} + S - Lh_{(L+1)} = S$. So if $S \geq Y$, either $L = m$, $S = 0$ and $Y \leq 0$, or w with $h_w = h_{(L+1)}$ has $g(h_w) = S \geq Y$. Conversely, if $Y \leq 0$ then $Y \leq S$, since $S \geq 0$; and if $g(h_w) \geq Y$ then $S \geq g(h_w) \geq Y$.

4 Applications

A community currency serves a community whose members sell to the public and buy from one another. Each application below has a contract of its own, which extends GCC and says who accepts the community's coins in payment; GCC says who issues them and what backs them. Under GC a coin is paid only to its issuer, and a swap takes a coin in return, so an application's contract lets a member sell community coins to the public for fiat and lets the public pay any member with them.

Ride hailing. A cooperative of drivers is the community and its drivers are its treasurers. A rider obtains driver-coins by swapping sovereign coins or paying fiat to a driver, and pays any driver with them; a driver who holds a driver-coin issued by another driver redeems it from its issuer or from any treasurer, in kind or for fiat, or pays with it in turn. The cooperative's contract, with its dispatching, its fares and its fines, is in preparation; under it every driver accepts driver-coins.

A merchants' association. The shops of a town form the community, their mutual credit the trade credit they extend each other, and their community coins the town's currency, which a customer buys from any shop and pays any shop with, as the businesses of Sardex pay each other in its unit [10,14]. The association's contract says that every shop accepts the association's coins.

A farmers' cooperative. The farms of a cooperative are its treasurers, and its coins are bought by the buyers of its produce, who pay any farm with them; a farm paid in another farm's coin redeems it in kind, for produce or coins the issuer holds, or for fiat. The cooperative's contract says that every farm accepts the cooperative's coins.

5 The App

The contracts GDC, GCC and BCC are checked syntactically grassroots and compiled down the abstraction cascade [7,11] to the mini-apps of the grassroots app; their screens follow.

6 Simulation

Communities of n treasurers, each extending every other k coins of mutual credit, whose treasurers fail at random, simulated in sGLP [22]: the coins in circulation and the coins left unredeemed, as the failures cross the fraction f.

7 Related Work

Community and complementary currencies. Sardex [10,14] and Sarafu [15] create liquidity from trust within a community, in units pegged by convention to the national currency and kept by an operator; personal-currency systems [3, 6,9] carry liquidity over trust edges as grassroots currencies do. A grassroots

community currency has no operator, is denominated and redeemable at par, and is backed by its treasurers' coins.

Grassroots currencies. Grassroots coins and bonds [18,21] are units of debt issued by any person, with mutual credit formed by their exchange; denominated grassroots currencies and the sovereign grassroots currency [23] add the fiat unit and the central bank. A community coin is issued by a treasurer of a community, denominated, and backed by other treasurers' coins.

Mutual credit, inside money and social collateral. Mutual credit is inside money [2,16], secured by social collateral [1]; the backing of a community coin is such credit, which its issuer may not give away while it is needed.

Backed money. Commercial bank money is pegged by redemption in cash and backed by reserves, insurance and the lender of last resort [16]; the note-issuing banks of free banking had the obligation and no backstop [17,25]; regulated stablecoins are backed one for one by reserves [8]. A community coin is backed by the solvency of the treasurers whose coins its issuer holds, whichever fraction f of them fail, by construction, with no reserve and no backstop.

Grassroots contracts and the cascade. A grassroots contract [7] is a social contract that runs, compiled through volition-guarded transactions [13] and the abstraction cascade [11] to Grassroots Logic Programs [20]; grassroots platforms are due to [19], their payment protocol is Grassroots Flash [12], a lost key or device is recovered by [4], and people are simulated by stochastic agents in sGLP [22].

Electronic records in law. Electronic-signature laws [5,24] make a signed record legally binding also in electronic form; a community coin is such a record, a debt of its issuer.

8 Conclusion

A community issues a denominated currency through its treasurers, under a grassroots contract and with no operator: a treasurer mints community coins only while the personal coins they hold of the other treasurers cover them after leaving out any fraction f of those treasurers, and a community coin stays redeemable for one unit of its denomination while at most a fraction f of the treasurers are insolvent or removed. The contracts are compiled to a working app, and simulated communities bear the theorem out. The treasurers are compensated by seigniorage, and interest, fees and loans are left to the instruments of grassroots bonds.

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A The Pragmas

The pragmas of GDC and GCC, which the apps built from them follow; GC's are in its paper [18].

GDC: Grassroots Denominated Coins, the pragmas

Pragmas

Minting: `Alice [Dmint] <denomination> <amount>`
Swap: `Alice [Swap] <amount given> <issuer wanted> <denomination wanted> <amount wanted>`
`Bob: Alice asks to swap coins with you.`
Payment: `Alice [Pay] <amount>`
Redemption: `Alice [Redeem] <denomination wanted> <issuer wanted>`

GCC: Grassroots Community Coins, the pragmas

Pragmas

Coins: the coins Alice holds.
Communities: the communities Alice is a treasurer of.
Community: `Alice [Start] <name> <denomination>`
Minting: `Alice [Cmint] <amount>`
Swap: `Alice [Swap] <amount given> <issuer wanted> <founder wanted> <community wanted> <denomination wanted> <amount wanted>`
`Bob: Alice asks to swap coins with you.`
Payment: `Alice [Pay] <amount>`
Redemption: `Alice [Redeem] <founder wanted> <community wanted> <denomination wanted> <issuer wanted>`
Treasurers: `Alice [Appoint] <person>`
`Bob: Alice asks you to become a treasurer of their community.`
`Carol: Alice asks to appoint Bob a treasurer of your community.`
`Alice [Remove]`
`Dave: Alice asks to remove Carol as a treasurer of your community.`
Coin Redemption: `Alice [Credeem] <founder wanted> <community wanted> <denomination wanted> <issuer wanted>`

BCC adds no act, so it has no pragmas of its own; GCC's stand.